

TuniCars+

Resilience Testing Report

Docker Swarm vs Kubernetes

Session ID

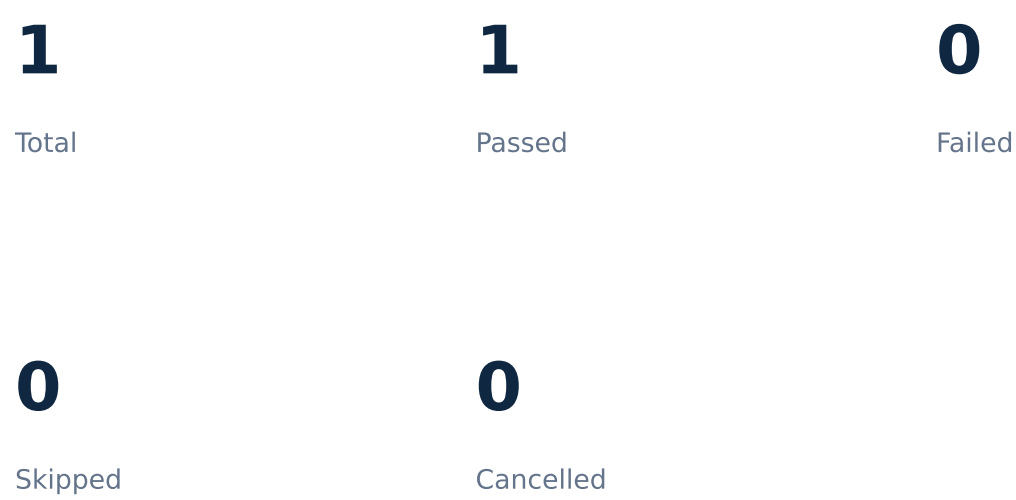
20260813_135332_745426

Date

2026-08-13T12:53:32.439186+00:00

Executive Summary

20260813_135332_745426



Overall measured availability: 61.29%

During this campaign, 1 of 1 scenarios passed, 0 failed and 0 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.

Test Environment

Architecture and technologies

Frontend → Backend → PostgreSQL

Docker Swarm environment

- Docker services
- FastAPI backend
- Next.js frontend
- PostgreSQL database

Kubernetes / Minikube environment

- Kubernetes Deployments and Services
- Minikube nodes
- FastAPI · Next.js · PostgreSQL

Test Scenarios

Scenario	kubernetes
memory	1

Full Results Table

Scenario	Platform	Detection	Recovery	Required	Success	Avg ms	P95 ms	HTTP Fail	Avail. %	Status	Skip Reason	Error Message
memory	kubernetes	63.25	12.266	True	True	1914.534	4026.552	24	61.29	PASS	N/A	N/A

kubernetes — memory

State: **Pass**
Detection Time (s): 63.25
Recovery Time (s): 12.266
HTTP Success: N/A
HTTP Failures: 24
Availability (%): 61.29
CPU (%): N/A
Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

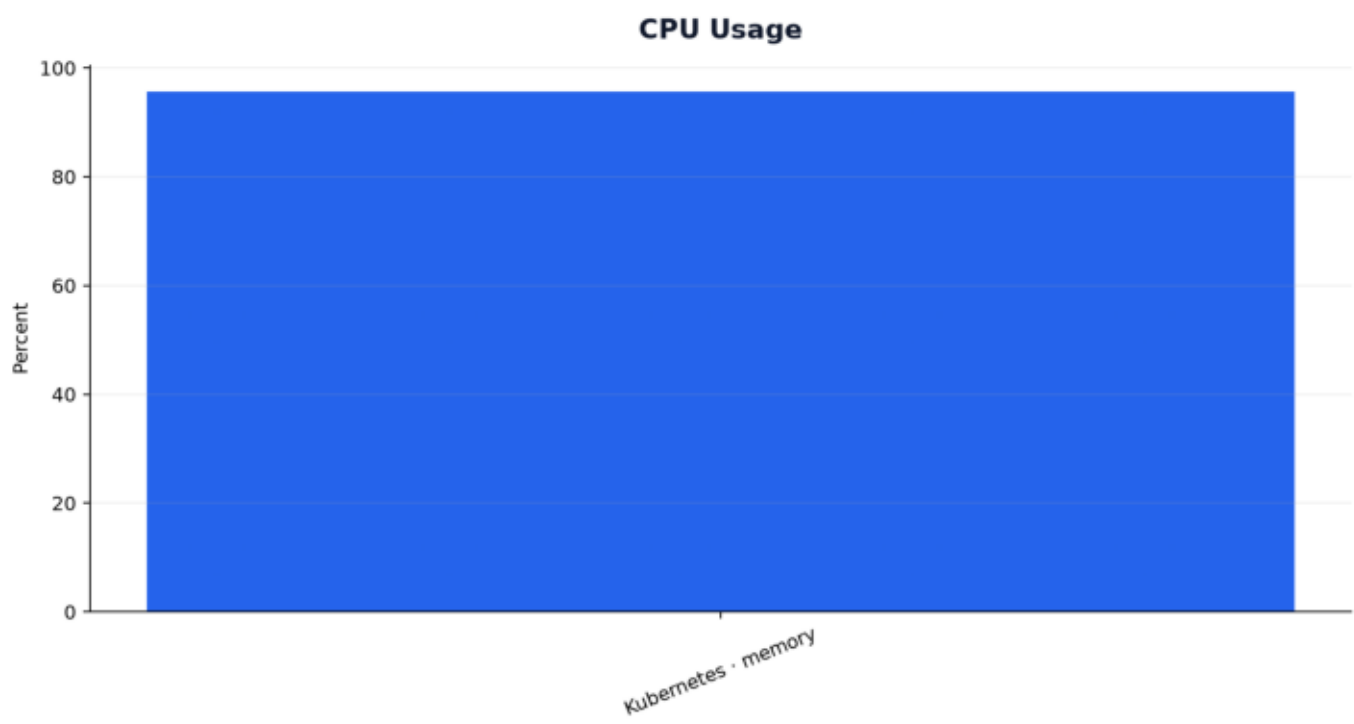
Availability

Current-session measured data



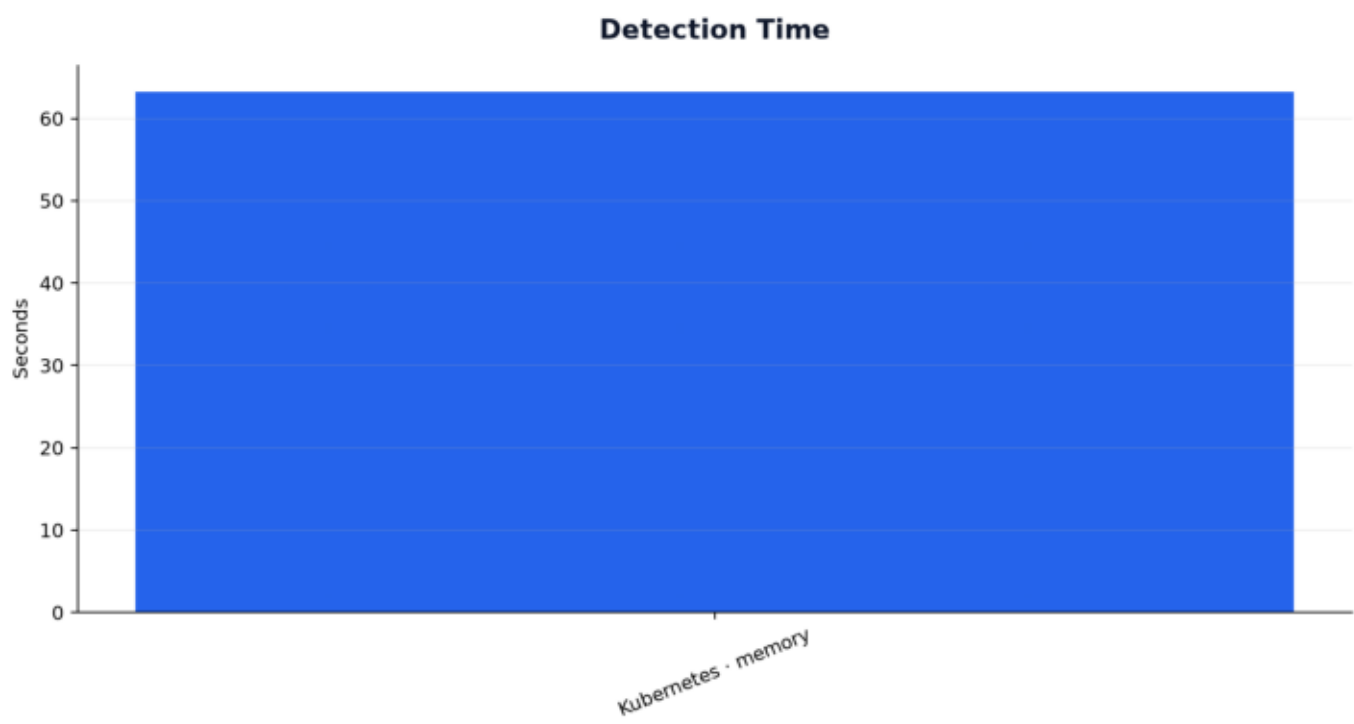
Cpu Usage

Current-session measured data



Detection Time

Current-session measured data



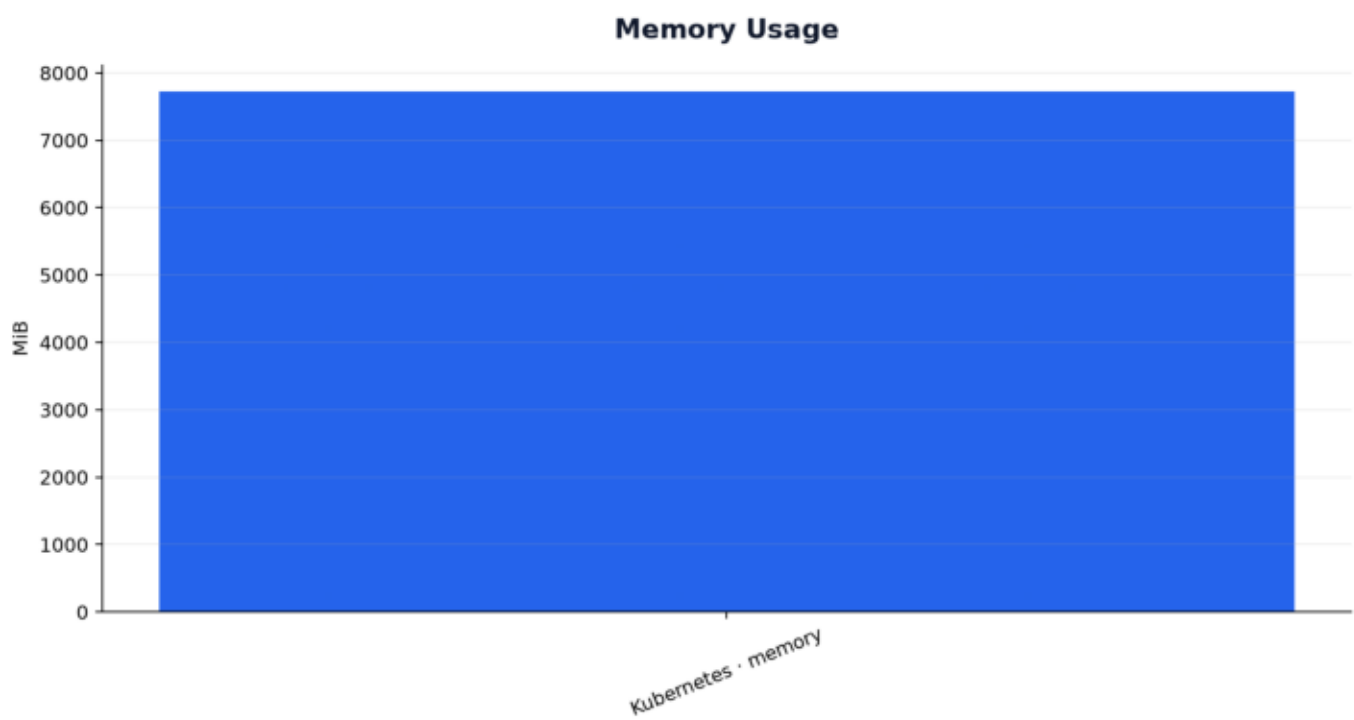
Http Failures

Current-session measured data



Memory Usage

Current-session measured data



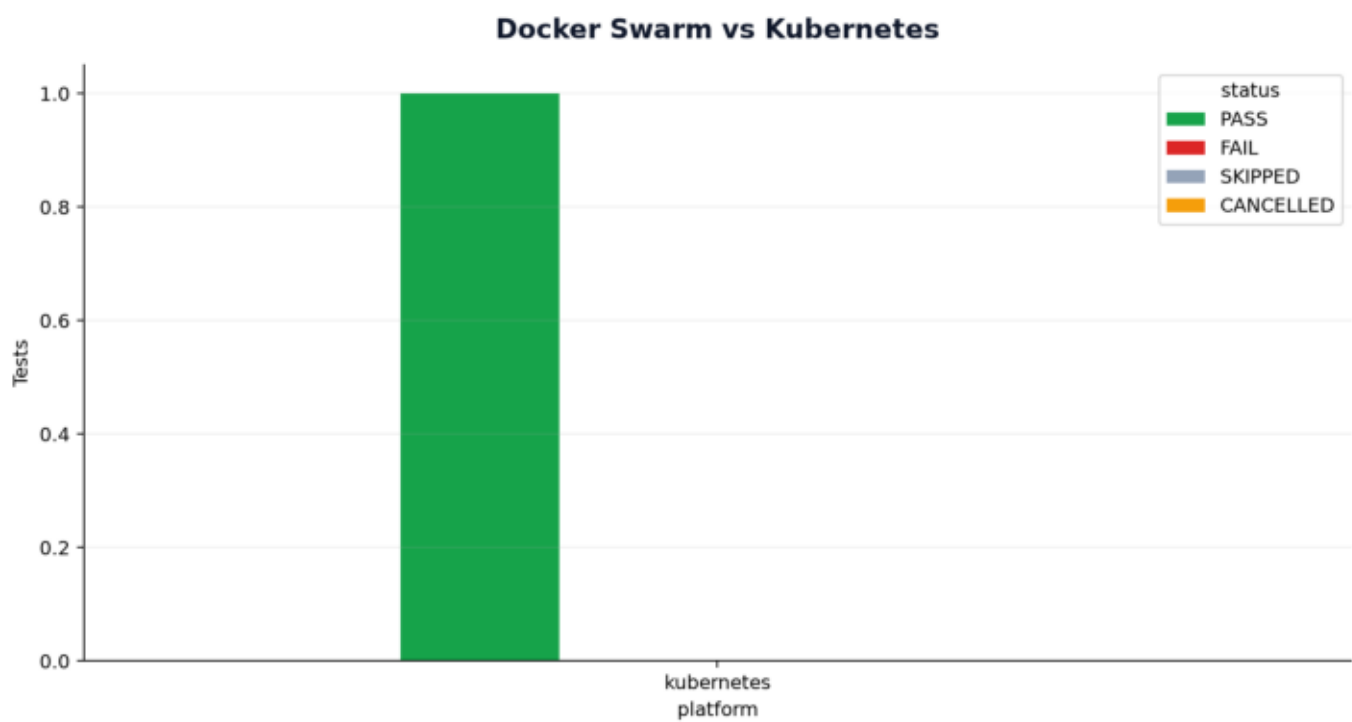
P95 Response Time

Current-session measured data



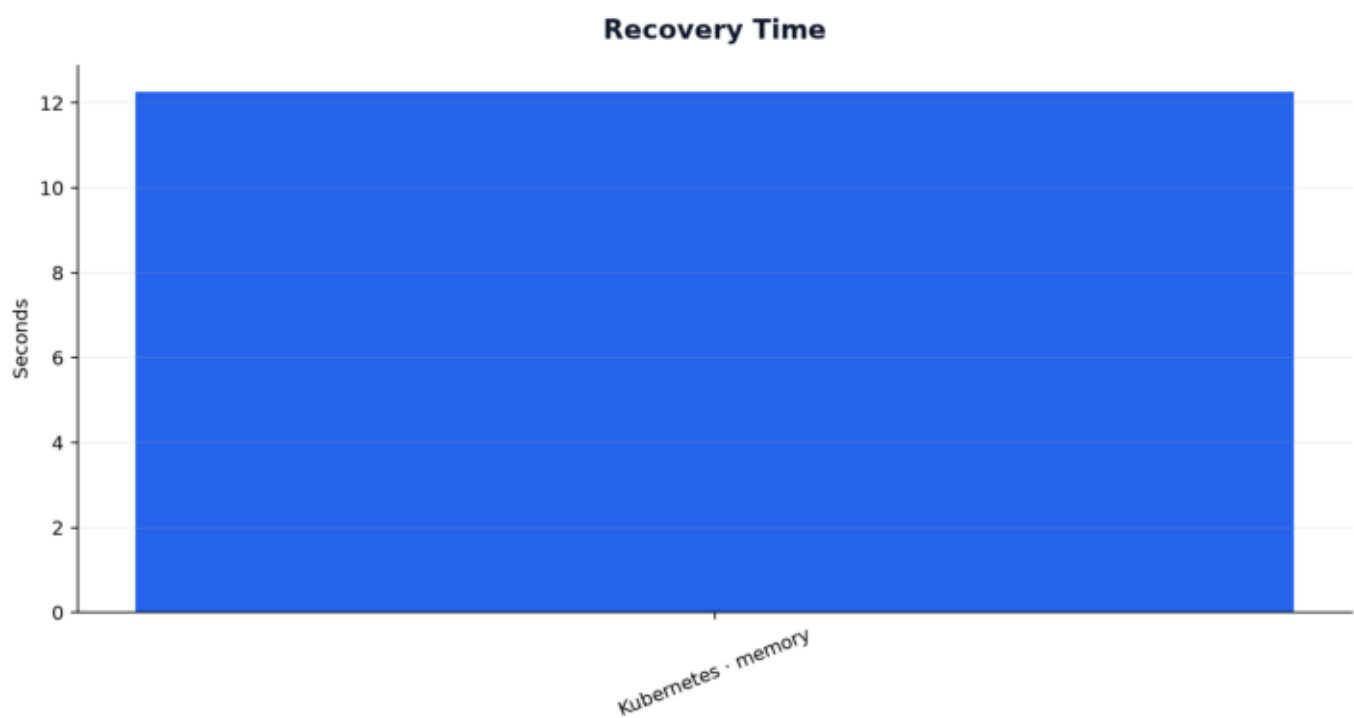
Platform Comparison

Current-session measured data



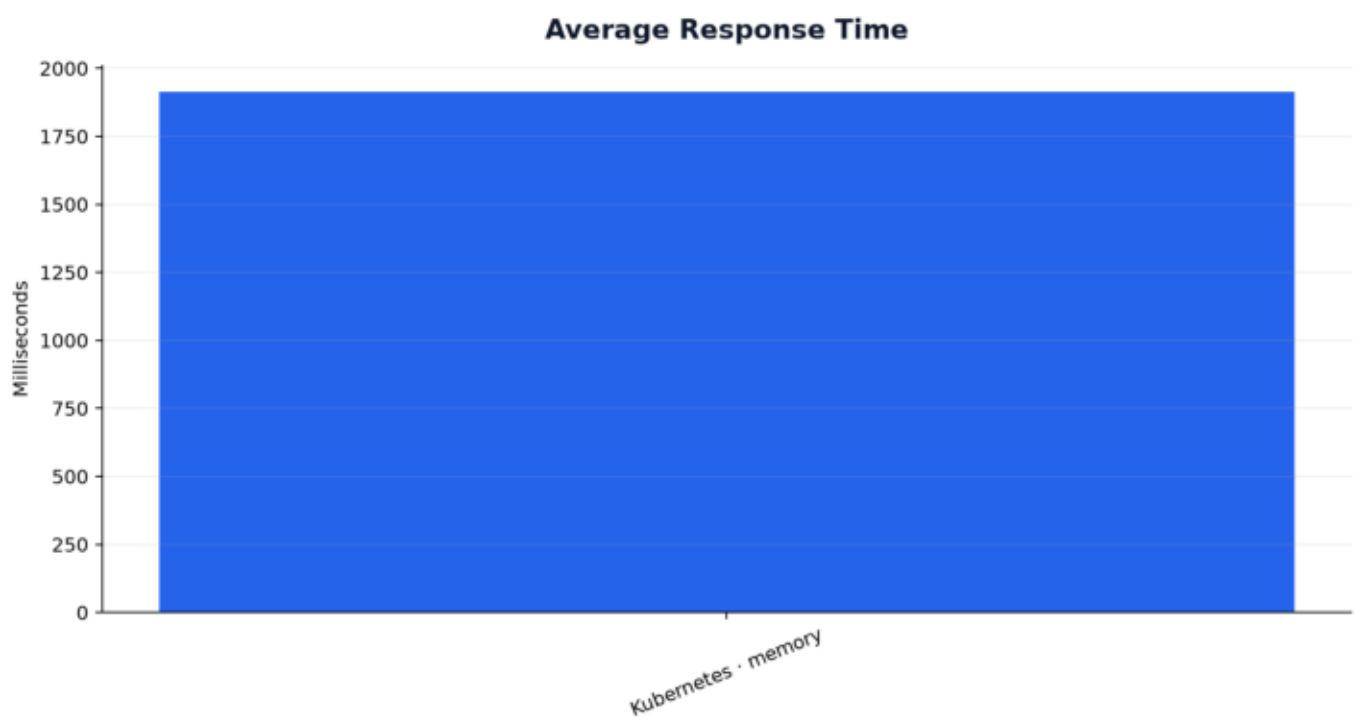
Recovery Time

Current-session measured data



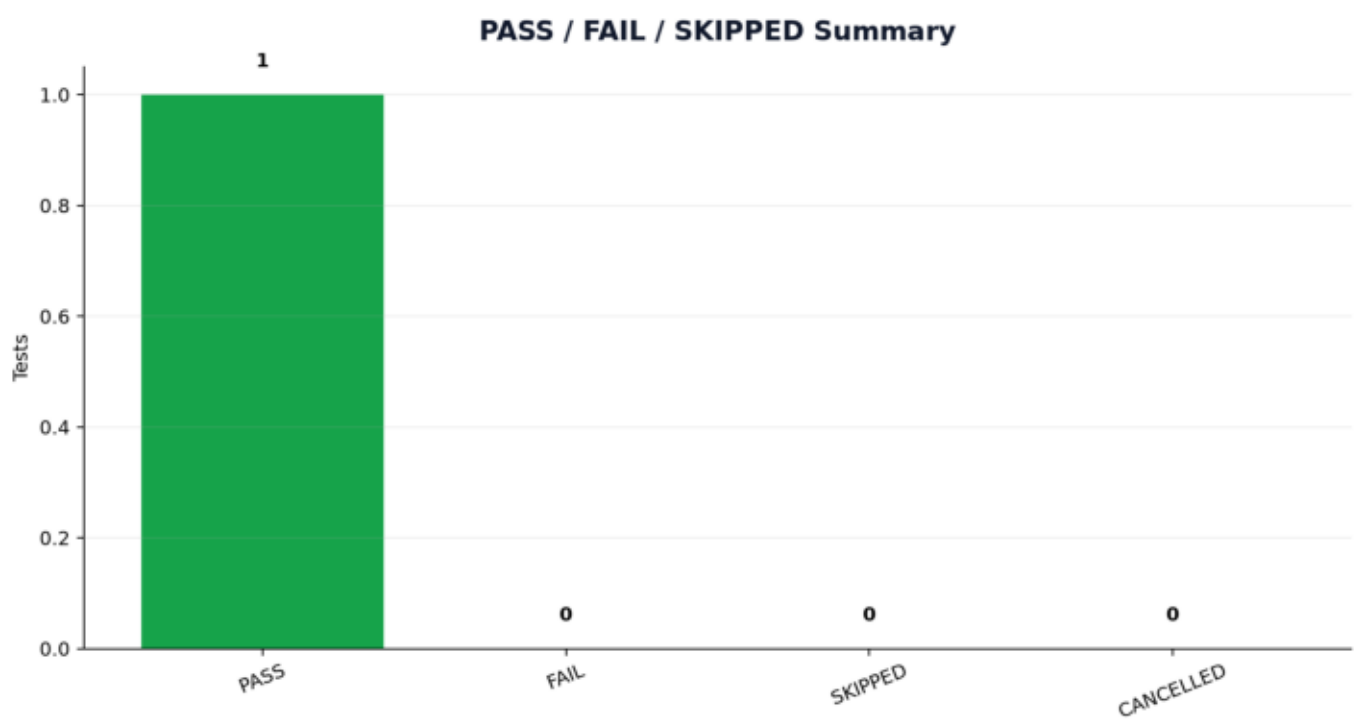
Response Time

Current-session measured data



Status Summary

Current-session measured data



Failures and Errors

Platform	Scenario	Status	Error
—	—	—	No failures recorded.

Skipped Tests

Platform	Scenario	Exact reason
—	—	No skipped tests recorded.

Swarm vs Kubernetes Comparison

Platform	PASS
kubernetes	1

Final Conclusion

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During this campaign, 1 of 1 scenarios passed, 0 failed and 0 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.